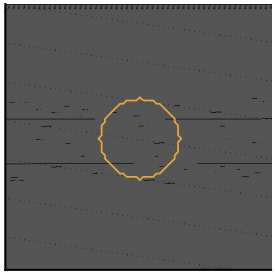
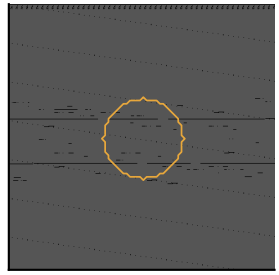
Controlled validation: signal patterns and edge cases

a Uniform



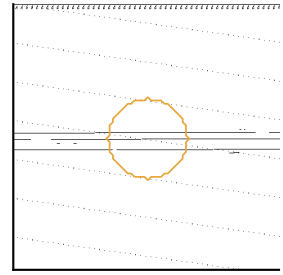
diff=0.00; enrich=1.00
CV=0.00; offset/r=0.00

b Bright ring, dark inner



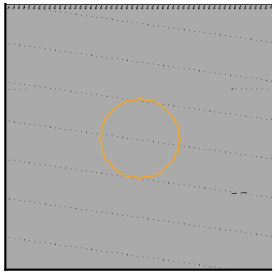
diff=10.00; enrich=2.98
CV=0.00; offset/r=0.00

c One-sided spot



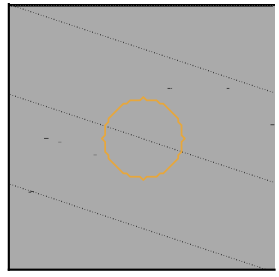
diff=1.51; enrich=1.61
CV=2.25; offset/r=0.81

d Bright neighbor, dark target



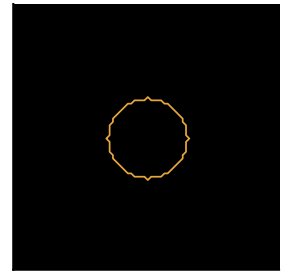
whole=0.00
contact diff=0.00

e Signal at contact side



whole=0.27
contact diff=12.00

f All dark

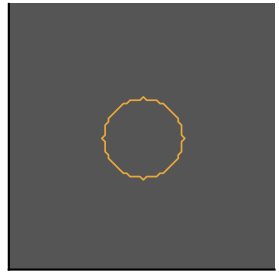


no_signal / MAD=0
Affected features: NaN

g Tiny instance

small_whole / small_inner
Affected features: NaN

h No local reference



reference_insufficient
Affected features: NaN

i Truncated at image border

truncated
Affected features: NaN

Orange contour: target instance. diff: edge-inner; enrich: area-adjusted enrichment; CV: sector signal variation.
Undefined localization and geometry are represented by NaN/QC; these are not cell-positivity rules.